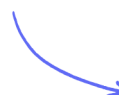


Structured Questions

Redox Chemistry & Acid-Base Titrations

Oxidation Numbers - Introduction / Oxidation & Reduction / Disproportionation Reactions / Ionic Equations / Acid-Base Titrations with Indicators

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Total Marks

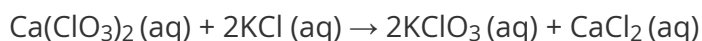
/22

- 1 (a) Potassium chlorate(V), KClO_3 , is a crystalline solid used in fireworks. It is produced by the Liebig Process in two stages.

Stage 1	Chlorine gas is passed through hot calcium hydroxide solution forming calcium chlorate(V), $\text{Ca}(\text{ClO}_3)_2$.
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Stage 2	Potassium chloride solution is then added to form potassium chlorate(V).
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The solution is heated to reduce its volume and then allowed to crystallise. The crystals are filtered off. The remaining filtrate is evaporated further to obtain more crystals.

- i) Write the **overall** equation for the Liebig Process. State symbols are not required.

(1)

- ii) Calculate the **overall** atom economy by mass for the production of potassium chlorate(V), KClO_3 , using your equation in (a)(i).

(3)

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(4 marks)

- (b) Explain the type of reaction that takes place in Stage 1 of the Liebig Process, using oxidation numbers.



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(3 marks)

(c) The crystals of potassium chlorate(V) formed also contain some halide ion impurities.

i) Describe a chemical test on a solution of these crystals to **confirm** that the impurities present are chloride ions rather than bromide ions. Include the expected results.

(3)

ii) 1.52 g of impure potassium chlorate(V), formed in the Liebig Process, was heated until the mass of solid remaining was constant at 1.02 g.

The reaction that occurred was



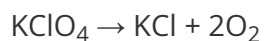
The impurities present did not decompose on heating.

Calculate the percentage purity of the sample. Give your answer to an appropriate number of significant figures.

(5)

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- (d) In fireworks, potassium chlorate(V) decomposes. This thermal decomposition takes place in two stages with a solid catalyst.



- i) Give the systematic name of KClO_4 .

(1)

- ii) A student investigated the role of the catalyst in this reaction.

Procedure

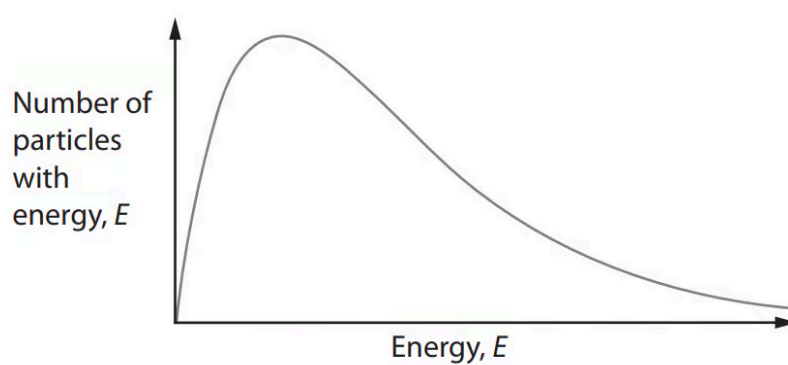
Step 1	Heat a sample of KClO_3 , in a test tube, with a known mass of insoluble catalyst until the mass remains constant.
Step 2	Mix the contents of the test tube with water.
Step 3	Filter the mixture and rinse with deionised water.
Step 4	Dry the remaining solid.
Step 5	Measure the mass of the dry solid.

Explain how each of the steps in this procedure is needed to show that the catalyst is **not** used up in this reaction.

(4)

(5 marks)

(e) Explain, using the diagram, how a catalyst speeds up a chemical reaction.



(2 marks)